

CAT Algebra – Mixed Practice Set 2 of 3

Topics: Polynomials · Quadratics · Binomial · Inequalities · Functions · GIF · Modulus · Logarithms · Progressions · Co-ordinate Geometry · Maxima & Minima

QUESTIONS

Q1.

[Functions]

$f(x)f(y)=f(xy)$ for all real x,y and $f(2)=4$. What is $f(1/2)$?

- (1) 0
- (2) $1/4$
- (3) $1/2$
- (4) 1

Q2.

[Functions]

$g(x+1)+g(x-1)=g(x)$ for all real x . Minimum p such that $g(x+p)=-g(x)$?

- (1) 2
- (2) 3
- (3) 5
- (4) 6

Q3.

[Greatest Integer Function]

If $[x]^2+4\{x\}=2x$, how many values of x are possible?

- (1) 1
- (2) 2
- (3) 3
- (4) 4

Q4.

[Greatest Integer Function]

Number of solutions of $[x]=x+1$:

- (1) 0
- (2) 1
- (3) 2
- (4) Infinite

Q5.

[Greatest Integer Function]

How many solutions does $11[y]+23\{y\}=250$ have?

- (1) 0
- (2) 1
- (3) 2
- (4) 3

Q6.

[Modulus]

If $7+3|x|=13$, find x .

- (1) -2.5

- (2) ± 6
- (3) ± 2
- (4) $\pm 2/7$

Q7.

[Modulus]

Range of real x satisfying $|2x+7| \geq |3x+9|$:

- (1) $[-7/2, -2]$
- (2) $(-\infty, -2]$
- (3) $(-\infty, -16/5]$
- (4) $(-\infty, -1]$

Q8.

[Modulus]

If $|3x|+7|x| \leq 60$, the value of x is:

- (1) $-7 < x < 7$
- (2) $x < 6$
- (3) $-6 < x < 6$
- (4) $-6 \leq x \leq 6$

Q9.

[Modulus]

How many integral values of x satisfy $||2x-19|-7| < 5$?

- (1) 13
- (2) 12
- (3) 11
- (4) 10

Q10.

[Logarithm]

How many digits in 2^{50} ? ($\log_{10} 2 = 0.301$)

- (1) 15
- (2) 16
- (3) 17
- (4) 18

Q11.

[Logarithm]

If $\log_y x = a \cdot \log_z y = b \cdot \log_x z = ab$, which pair (a, b) is NOT possible?

- (1) $(-1/2, 2)$
- (2) $(1, 1)$
- (3) $(0.4, 2.5)$
- (4) $(2, 2)$

Q12.

[Progressions]

Series 2, 5, 8, ... (50 terms). 15th term in descending order is:

- (1) 104
- (2) 107
- (3) 110
- (4) 113

Q13.

[Progressions]

In infinite GP, each term = twice sum of all following terms. Sum of first two terms is 12. Sum of entire GP:

- (1) $9/2$
- (2) $27/2$
- (3) $88/7$
- (4) 15

Q14.

[Progressions]

Sum of infinite GP=14; sum of cubes=392. First term is:

- (1) -14
- (2) 10
- (3) 7
- (4) -5

Q15.

[Progressions]

For $a_1=81.33$, $a_2=-19$, $a_j=a_{j-1}-a_{j-2}$, sum of first 6002 terms is:

- (1) -100.33
- (2) -30.00
- (3) 62.33
- (4) 119.33

ANSWER KEY

Q1: (2) $1/4$ **Q2:** (2) 3 **Q3:** (3) 3 **Q4:** (1) 0 **Q5:** (3) 2
Q6: (3) ± 2 **Q7:** (1) $[-7/2, -2]$ **Q8:** (4) $-6 \leq x \leq 6$ **Q9:** (4) 10 **Q10:** (2) 16
Q11: (4) (2,2) **Q12:** (2) 107 **Q13:** (2) $27/2$ **Q14:** (3) 7 **Q15:** (3) 62.33

DETAILED SOLUTIONS

Solution Q1 — [Functions]

Answer: (2) $1/4$

$f(2) \cdot f(1/2) = f(1) = 1$. So $f(1/2) = 1/4$.

Solution Q2 — [Functions]

Answer: (2) 3

From recurrences: $g(x+3) = -g(x)$. So minimum $p=3$.

Solution Q3 — [Greatest Integer Function]

Answer: (3) 3

$f = (2I - I^2)/2$ must satisfy $0 \leq f < 1$. Valid: $I=0$ ($x=0$), $I=1$ ($x=1.5$), $I=2$ ($x=2$). Three solutions.

Solution Q4 — [Greatest Integer Function]

Answer: (1) 0

Integer x : $x = x+1$ impossible. Non-integer $x = I + f$: $I = I + f + 1 \rightarrow f = -1$, impossible. No solutions.

Solution Q5 — [Greatest Integer Function]

Answer: (3) 2

$11[y]$ in $(227, 250]$: $[y] = 21$ or 22 . Each gives a valid $\{y\}$. Two solutions.

Solution Q6 — [Modulus]

Answer: (3) ± 2

$3|x| = 6 \rightarrow |x| = 2 \rightarrow x = \pm 2$.

Solution Q7 — [Modulus]

Answer: (1) $[-7/2, -2]$

Squaring and solving: $(5x+16)(x+2) \leq 0 \rightarrow -16/5 \leq x \leq -2$, with check giving $[-7/2, -2]$.

Solution Q8 — [Modulus]

Answer: (4) $-6 \leq x \leq 6$

For $x \geq 0$: $10x \leq 60 \rightarrow x \leq 6$. For $x < 0$: $-10x \leq 60 \rightarrow x \geq -6$. So $-6 \leq x \leq 6$.

Solution Q9 — [Modulus]

Answer: (4) 10

Gives $2 < |2x-19| < 12$. Case 1: $21 < 2x < 31 \rightarrow 5$ integers (11–15). Case 2: $7 < 2x < 17 \rightarrow 5$ integers (4–8). Total: 10.

Solution Q10 — [Logarithm]

Answer: (2) 16

$\log_{10}(2^{50}) = 15.05$. Characteristic = 15 $\rightarrow$ 16 digits.

Solution Q11 — [Logarithm]

Answer: (4) (2,2)

The constraint requires $ab=1$. For (2,2): $ab=4 \neq 1 \rightarrow$ impossible.

Solution Q12 — [Progressions]

Answer: (2) 107

15th from end = 36th from start. $T_{36} = 2 + 35 \times 3 = 107$.

Solution Q13 — [Progressions]

Answer: (2) $27/2$

$r=1/3$, $a=9$. $\text{Sum} = a/(1-r) = 27/2$.

Solution Q14 — [Progressions]

Answer: (3) 7

$a/(1-r)=14$, $a^3/(1-r^3)=392$. Solving: $r=1/2$, $a=7$.

Solution Q15 — [Progressions]

Answer: (3) 62.33

Period 6, zero sum per cycle. $6002 = 6 \times 1000 + 2$, so $\text{sum} = a_1 + a_2 = 62.33$.
